Arkashine School Innovation Lab (SIL)

Summer Innovation Program (72 Training Hours) • Complete Site Flow & Architecture

1. Three-Level Operational Hierarchy & Navigation Flow

The web application provides 5 distinct role-scoped views with zero dead-ends and max 2-click reachability:

Role & Scope	User Persona	Screens & Primary Flow
1. Arkashine HQ Super Admin	Dr. Vivek Saxena Managing Director	• **Overview**: 4 KPI cards, school deployment hub, Needs Attention list. • **Schools & Partners**: 18-point Eligibility gating (Eligible / Conditional / Rejected). • **Program Heat Map**: 28-day compliance across all schools. • **Impact & ESG**: Environmental (pH, OC, samples), Social (girls 52%), Governance. • **Certificates**: Bulk validation and digital issuance.
2. Enabling Partner Implementation	Dr. Sunita Deshmukh GreenHorizon AgriEd	• **Overview**: RAG health status cards for assigned schools. • **My Schools & Lab Setup**: Sensor hardware & consumables tracking. • **Student Selection**: 4-stage funnel (120 Registered → 60 Quiz → 30 Assignment → 18 Cohort). • **Outreach & Support**: Support tickets directly to Arkashine HQ.
3. School Principal School Head	Fr. Thomas Matthew St. Xavier's Senior Sec.	• **My School**: 4-card scorecard (Hours, Attendance, Tests, Farmers) + 4-module progress. • **Daily Log**: Chronological timeline of faculty updates with endorsement remarks. • **Students & Risk**: Monitors students at-risk (<80% attendance). • **Certificates**: Principal verification and endorsement.
4. Faculty Coordinator Field Teacher	Dr. Anjali Patil Science Faculty	• **Today's Update**: Quick-log engine (<3 min, phone-optimized) with auto-saving draft. • **Attendance**: 1-tap toggle (Present ■ → Late ■ → Absent ■). • **Soil Lab (SoiLENZ)**: Logs tests (pH, EC, Organic Carbon, N, P, K). • **Outreach**: Records farmer meetings and village campaigns.
5. Student Portal Learner View	Ananya Sharma Class X-A, Group 1 Lead	• **My Soil Journey**: 8-step precision soil workflow progression. • **Modules & Quizzes**: Theory (Mod 1) & Practicals (Mod 2, 3, 4). • **Badges**: Unlocks Sampler, Tester, Reporter, Farmer Friend. • **Certificate**: Live preview with tamper-proof QR code verification.

2. Curriculum Modules (Theory vs. Practical Breakdown)

The 72-hour curriculum is structured so students master fundamental science and practical field techniques:

Module	Type	Title & Hours	Key Focus & Hands-on Activities
Module 1	**Theory** ■	**Introduction to Agriculture & Soil Science** 18 Training Hours	Soil physics, chemistry (pH, EC, CEC), soil horizons (O, A, B, C), N-P-K nutrient cycles, and regenerative soil carbon dynamics. Assessment: Comprehensive Theory Quiz.
Module 2	**Practical** ■	**Soil Sample Collection & Preparation** 18 Training Hours	Auger sampling at 0-15cm furrow depth, zig-zag traverses, GPS geotagging, composite quartering, shade drying, and 2mm sieving. Assessment: Quiz + Field Auger Assignment.
Module 3	**Practical** ■	**Soil Testing using SoiLENZ Device** 18 Training Hours	Handheld SoiLENZ spectroscopic sensor operation, two-point buffer calibration (pH 4.0/7.0), quartz cuvette loading, and rapid 90-sec testing. Assessment: Quiz + SoiLENZ Challenge.

Module 4	**Practical ■**	**Soil Health Reports & Farmer Advisory** 18 Training Hours	Generating Soil Health Cards, calculating customized fertilizer dosage, and student-led vernacular advisory meetings with village farmers. Assessment: Quiz + Capstone Project.

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3. Student 8-Step Precision Soil Testing Lifecycle

Students follow an 8-stage hands-on workflow from rural field sampling to farmer advisory:

Step 1: Collect Auger core 0-15cm	**Step 2: Prepare** Air dry & 2mm sieve	**Step 3: Test** SoiLENZ 90s scan	**Step 4: Record** Telemetry sync
Step 5: Analyze N-P-K & OC balance	**Step 6: Interpret** Soil Health Index	**Step 7: Report** Print Health Card	**Step 8: Apply** Farmer advisory

4. Faculty Coordinator Daily Attendance & Storing Engine

Designed for quick phone-browser entry (<3 minutes) by school teachers:

1. Session Format	Select Module (1-4) and Session Type (Theory vs Practical). Stepper sets hours (2.5h default).
2. 1-Tap Attendance	Large student tap cards toggle: Present ■ → Late ■ → Absent ■ (or 'All Present' button). Drafts auto-save locally so data is never lost.
3. Lab Telemetry	Input count of SoiLENZ tests performed, farmers reached, and attach lab photo evidence.
4. Submit & Store	Permanently stored in local database. Instantly updates student hours, unlocks badges, and turns the school's calendar heat map green for Principal and Arkashine HQ.

5. Strict Quality Gating & Certification Pipeline

Gate 1: School Eligibility Approval
• 18 verification points across Space, Infrastructure, Safety, Faculty, Students (80+ registrations), and signed MoU.
• Requires score ≥85% with 0 critical flags to move school from 'Pending' to 'Enrolled' status.

Gate 2: Soil Partner Certification Gate
• Strict prerequisites: 1) ≥80% student attendance (≥58 of 72 training hours), 2) Pass all 4 modules, 3) Complete SoiLENZ testing practical, 4) Complete farmer advisory.
• Generates official A4 Certificate with leaf guilloche border and tamper-proof verification QR code.

6. Live Color Theme Customizer (For Demo Presentations)

A dedicated **Theme Colors** ■ button in the top navigation bar allows 1-click theme switching during client meetings:
• ■ **Forest Emerald** (Nature / Agritech) • ■ **Deep Navy & Sky** (Corporate Ed-Tech) • ■ **Warm Earth & Amber** (Soil & Farming)
• ■ **Modern Ocean Teal** (Sustainability) • ■ **Royal Purple** (Science & Innovation) • ■ **Minimal Slate** (Clean Monochrome)
Clicking 'Fine Tune' opens custom color pickers for instant real-time live preview.